

Equivalence of the Interband Absorption Integral with Rosei's Energy-Distributed Joint Density of States

A step-by-step reconciliation at the gold X and L critical points

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Abstract. We carry the interband contribution to $\varepsilon_2(\hbar\omega, T)$ of gold from Fermi's golden rule down to a one-dimensional energy integral, retaining every $\hbar^2/2m$ factor and the full non-equilibrium occupation $f(E_l) [1 - f(E_u)]$. We then place this reduction *line by line* beside the energy-distributed joint-density-of-states (EDJDOS) expression of Guerrisi, Rosei & Winsemius, *Phys. Rev. B* **12**, 557 (1975) [Eqs. (1)–(9)]. The two integrands are shown to be the *same function of* (E, ω) : they share the identical shape-carrying factor $k_{\parallel}^{-1}(E, \omega)$ and the identical integration limits, and differ only by (i) a multiplicative constant $16\pi^3 \mathcal{F}_{l \rightarrow u} / \hbar^2$ that is independent of E and ω and is absorbed into the overall fitted scale, and (ii) the valence-band-full approximation $f(E_l) \rightarrow 1$, under which our statistical factor collapses to Rosei's $[1 - f(E)]$. Finally we examine two of Rosei's printed display equations, (6) and (8), and show on dimensional, geometric, and internal-consistency grounds that the corrected upper integration limit must carry *perpendicular*, not parallel, effective masses.

1 Setup, notation, and the claim

Both treatments describe the same physical process: a vertical (direct, $\mathbf{k}_u = \mathbf{k}_l \equiv \mathbf{k}$) transition from the filled d -derived band l to the sp band u near the X point of gold, under rotational symmetry about the ΓX axis. Rosei's quadratic dispersions, measured from the Fermi level, are his Eqs. (1) and (2):

$$E = \hbar\omega_u = \hbar\omega_{X_6^-} + \frac{\hbar^2 k_{\perp}^2}{2m_{u\perp}} - \frac{\hbar^2 k_{\parallel}^2}{2m_{u\parallel}}, \quad (1) \quad (\text{saddle: } +k_{\perp}^2, -k_{\parallel}^2) \quad (1)$$

$$E - \hbar\omega = \hbar\omega_l = -\hbar\omega_{X_7^+} - \frac{\hbar^2 k_{\perp}^2}{2m_{l\perp}} - \frac{\hbar^2 k_{\parallel}^2}{2m_{l\parallel}}, \quad (2) \quad (\text{maximum: } -k_{\perp}^2, -k_{\parallel}^2) \quad (2)$$

Throughout we use the compact curvature coefficients

$$A_i = \frac{\hbar^2}{2m_{i\perp}}, \quad B_i = \frac{\hbar^2}{2m_{i\parallel}}, \quad i = l, u, \quad (3)$$

so that the interband (transition) energy and the vertical gap are

$$\Omega_{lu}(\mathbf{k}) \equiv \hbar\omega_u - \hbar\omega_l = \mathcal{E}_g^X + \bar{A} k_{\perp}^2 + \bar{B} k_{\parallel}^2, \quad \bar{A} = A_u + A_l, \quad \bar{B} = B_l - B_u, \quad \mathcal{E}_g^X = \hbar\omega_{X_6^-} + \hbar\omega_{X_7^+}. \quad (4)$$

The minus sign on $k_{\parallel}^2$ in (1) makes u a saddle, so Ω_{lu} is a saddle in $\mathbf{k}$ and the constant-energy-difference surface (CEDS) $\Omega_{lu} = \hbar\omega$ is an *open* hyperboloid; this is the origin of the sub-gap tail treated in section 6. The companion L point differs only in the sign of the upper-band $k_{\parallel}^2$ curvature and is collected in section 6.

The two integrands to be reconciled. The object of this note is to prove the identity boxed below.

Claim

The absorption integrand obtained here by an explicit change of variables,

$$\varepsilon''(\omega) \propto \int f(E - \hbar\omega) [1 - f(E)] \mathcal{J}(E) dE, \quad \mathcal{J}(E) = \frac{2\pi k_{\perp}}{|\det J|} \Big|_{\Delta=\hbar\omega},$$

and Rosei's EDJDOS integrand [Eqs. (4), (7)],

$$\varepsilon_2^{\text{Rosei}}(\omega) \propto \int [1 - f(E)] D_{l \rightarrow u}(E, \hbar\omega) dE, \quad D_{l \rightarrow u}(E, \hbar\omega) = (8\pi^2 \hbar^2)^{-1} \mathcal{F}_{l \rightarrow u} k_{\parallel}^{-1},$$

describe the *same* spectral function. Explicitly,

$$\boxed{\mathcal{J}(E) = \underbrace{\frac{16\pi^3 \mathcal{F}_{l \rightarrow u}}{\hbar^2}}_{\text{const. in } E, \omega} D_{l \rightarrow u}(E, \hbar\omega)} \quad \text{and} \quad f(E_l) [1 - f(E_u)] \xrightarrow{f(E_l) \rightarrow 1} [1 - f(E)].$$

2 The two reductions, line by line

We first display the logical skeletons in parallel, then carry out the one nontrivial common step — the Jacobian — in section 3.

STEP 0 • Starting point	
$\varepsilon'' \propto \iiint f(E_l) [1 - f(E_u)] \delta(\Omega_{lu} - \hbar\omega) d^3k$	$\varepsilon_2 = \frac{8\pi^2 e^2 \hbar^4}{3m^2 (\hbar\omega)^2} \sum_P P_P ^2 g_P \quad (9)$
golden rule, explicit occupation	$g_P = N_P I_{l \rightarrow u}$, with $I_{l \rightarrow u}$ from (7)
STEP 1 • Exploit cylindrical symmetry	
$d^3k \rightarrow 2\pi k_{\perp} dk_{\perp} dk_{\parallel}$	$d^3k \rightarrow 2\pi k_{\perp} dk_{\perp} dk_{\parallel}$
same measure	same measure (Ref. 10 procedure)
STEP 2 • Change variables $(k_{\perp}, k_{\parallel}) \rightarrow (E, \cdot)$	
$E \equiv E_u, \quad \Delta \equiv E_u - E_l$	$E \equiv E_u, \quad \Omega \equiv E_u - E_l$
$\Delta = \mathcal{E}_g^X + \bar{A} k_{\perp}^2 + \bar{B} k_{\parallel}^2$	identical map; $\Omega_{lu} = 0$ is (3)
STEP 3 • Jacobian of the map (see §3)	
$ \det J = 4k_{\perp} k_{\parallel} \mathcal{D}$	$\mathcal{D} = \left(\frac{\hbar^2}{2}\right)^2 \mathcal{F}_{l \rightarrow u}^{-2}$
$\mathcal{D} = A_u B_l + A_l B_u$ (mass constant)	folded into $\mathcal{F}_{l \rightarrow u}$, Eq. (5)
STEP 4 • Collapse the energy-conserving δ	
$\mathcal{J}(E) = \frac{2\pi k_{\perp}}{4k_{\perp} k_{\parallel} \mathcal{D}} = \frac{2\pi \mathcal{F}_{l \rightarrow u}^2}{\hbar^4 k_{\parallel}}$	$D_{l \rightarrow u} = \frac{\mathcal{F}_{l \rightarrow u}}{8\pi^2 \hbar^2 k_{\parallel}} \quad (4)$
shape carried entirely by $k_{\parallel}^{-1}(E, \omega)$	shape carried entirely by $k_{\parallel}^{-1}(E, \omega)$

STEP 5 • Statistical weight

$$f(E - \hbar\omega) [1 - f(E)]$$

general / non-equilibrium-ready

$$[1 - f(E)] \quad ((7))$$

$f(E_l) \rightarrow 1$ limit (filled d band)

Reading down the two columns, every line is either identical or differs by a constant prefactor. Only Steps 3–5 require argument; we supply each in turn.

3 Step 3 — the Jacobian is the band-curvature determinant

With $E = E_u$ from (1) and $\Delta = \Omega_{lu}$ from (4), the Jacobian matrix of (E, Δ) with respect to $(k_\perp, k_\parallel)$ is

$$J = \begin{pmatrix} \partial_{k_\perp} E & \partial_{k_\parallel} E \\ \partial_{k_\perp} \Delta & \partial_{k_\parallel} \Delta \end{pmatrix} = \begin{pmatrix} 2A_u k_\perp & -2B_u k_\parallel \\ 2\bar{A} k_\perp & 2\bar{B} k_\parallel \end{pmatrix}, \quad (5)$$

whose determinant factors the momenta out cleanly:

$$\det J = 4k_\perp k_\parallel (A_u \bar{B} + B_u \bar{A}) = 4k_\perp k_\parallel [A_u (B_l - B_u) + B_u (A_u + A_l)] = 4k_\perp k_\parallel (A_u B_l + A_l B_u). \quad (6)$$

Hence

$$|\det J| = 4k_\perp k_\parallel \mathcal{D}, \quad \boxed{\mathcal{D} \equiv A_u B_l + A_l B_u} \quad (7)$$

is a pure effective-mass constant — no E , ω , or k dependence. The factor of $k_\perp$ in the numerator measure $2\pi k_\perp$ cancels the $k_\perp$ in $|\det J|$, leaving the entire spectral shape in the single surviving factor $k_\parallel^{-1}$. Expanding (7) with (3),

$$\mathcal{D} = \left(\frac{\hbar^2}{2}\right)^2 \left(\frac{1}{m_{u\perp} m_{l\parallel}} + \frac{1}{m_{l\perp} m_{u\parallel}} \right) = \left(\frac{\hbar^2}{2}\right)^2 \frac{m_{l\perp} m_{u\parallel} + m_{l\parallel} m_{u\perp}}{m_{l\perp} m_{l\parallel} m_{u\perp} m_{u\parallel}}. \quad (8)$$

Comparing the mass ratio in (8) with Rosei's definition of $\mathcal{F}_{l \rightarrow u}$ [Eq. (5)],

$$\mathcal{F}_{l \rightarrow u} = \left(\frac{m_{l\perp} m_{u\parallel} + m_{l\parallel} m_{u\perp}}{m_{l\perp} m_{l\parallel} m_{u\perp} m_{u\parallel}} \right)^{-1/2} \implies \boxed{\mathcal{D} = \left(\frac{\hbar^2}{2}\right)^2 \mathcal{F}_{l \rightarrow u}^{-2}}. \quad (9)$$

Equation (9) is the bridge between the two notations: the determinant that my change of variables produces is precisely the inverse square of the constant Rosei defines. The two derivations have met.

4 Step 4 — the kernels coincide up to a constant

Insert (9) into the kernel of Step 4. From $|\det J| = 4k_\perp k_\parallel \mathcal{D}$ and $\mathcal{D} = (\hbar^2/2)^2 \mathcal{F}_{l \rightarrow u}^{-2}$,

$$\mathcal{J}(E) = \frac{2\pi k_\perp}{4k_\perp k_\parallel \mathcal{D}} = \frac{\pi}{2k_\parallel} \mathcal{D}^{-1} = \frac{\pi}{2k_\parallel} \left(\frac{\hbar^2}{2}\right)^{-2} \mathcal{F}_{l \rightarrow u}^2 = \frac{2\pi \mathcal{F}_{l \rightarrow u}^2}{\hbar^4 k_\parallel}. \quad (10)$$

Dividing by Rosei's $D_{l \rightarrow u} = \mathcal{F}_{l \rightarrow u} (8\pi^2 \hbar^2 k_\parallel)^{-1}$ [Eq. (4)], the $k_\parallel^{-1}$ cancels exactly and the ratio is constant:

$$\frac{\mathcal{J}(E)}{D_{l \rightarrow u}(E, \hbar\omega)} = \frac{2\pi \mathcal{F}_{l \rightarrow u}^2 / (\hbar^4 k_\parallel)}{\mathcal{F}_{l \rightarrow u} / (8\pi^2 \hbar^2 k_\parallel)} = 2\pi \mathcal{F}_{l \rightarrow u}^2 \cdot \frac{8\pi^2 \hbar^2}{\hbar^4 \mathcal{F}_{l \rightarrow u}} = \frac{16\pi^3 \mathcal{F}_{l \rightarrow u}}{\hbar^2}. \quad (11)$$

Identity of the kernels

$$\mathcal{J}(E) = \frac{16\pi^3 \mathcal{F}_{l \rightarrow u}}{\hbar^2} D_{l \rightarrow u}(E, \hbar\omega), \quad \text{with} \quad \frac{16\pi^3 \mathcal{F}_{l \rightarrow u}}{\hbar^2} \text{ independent of } E \text{ and } \omega.$$

The only carrier of spectral shape, $k_{\parallel}^{-1}(E, \omega)$, is *identical* in both; the integrands differ by a global multiplicative constant and nothing else.

4.1 Why the constant is not a discrepancy but a units bridge

The prefactor $16\pi^3 \mathcal{F}_{l \rightarrow u} / \hbar^2$ is exactly what reconciles the apparent “missing factor” between the bare EDJDOS and Rosei’s $D_{l \rightarrow u}$. Since $\mathcal{F}_{l \rightarrow u}^{-2}$ has dimensions $(\text{mass})^{-2}$, $\mathcal{F}_{l \rightarrow u}$ carries one power of mass, and

$$\left[\frac{\mathcal{F}_{l \rightarrow u}}{\hbar^2} \right] = \frac{\text{kg}}{(\text{J s})^2} = \frac{1}{\text{J m}^2} = \left[\frac{2m}{\hbar^2} \right]. \quad (12)$$

This is precisely the $\text{J}^{-1} \text{m}^{-2}$ by which Rosei’s $D_{l \rightarrow u}$ differs from a fully normalized energy-distributed JDOS (dimensions $[\text{vol}]^{-1} [\text{energy}]^{-2} = \text{m}^{-3} \text{J}^{-2}$). In words:

- Our $\mathcal{J} \propto \mathcal{F}_{l \rightarrow u}^2 / k_{\parallel}$ carries the dimensionally complete coefficient because every $\hbar^2 / 2m$ was retained explicitly through the Jacobian (7).
- Rosei pushed *one* power of $2m / \hbar^2$ into his definition of $\mathcal{F}_{l \rightarrow u}$ (leaving $\mathcal{F}_{l \rightarrow u}^1$ in Eq. (4)) and the rest into the overall normalization in front of ε_2 in Eq. (9). His $D_{l \rightarrow u}$ is therefore a *rescaled* EDJDOS, not the bare one.

Because $16\pi^3 \mathcal{F}_{l \rightarrow u} / \hbar^2$ is constant across the band, it is absorbed without trace into the single overall amplitude of ε_2 — the strengths S_X, S_L of Rosei’s Eq. (10) (the Scale parameter of the fit). **No spectral information is created, lost, or shifted.**

5 Step 5 — the statistical factors agree

The full microscopic weight of a vertical transition is the product “initial filled $\times$ final empty,”

$$f(E_l) [1 - f(E_u)] = f(E - \hbar\omega) [1 - f(E)], \quad (13)$$

where the second equality uses $E_l = E - \hbar\omega$ from the change of variables. For ordinary absorption the initial state l is the d -derived band lying well below the Fermi level: at the energies that contribute, $E_l \simeq -\hbar\omega_{X_7^+} \ll E_F$, so

$$f(E - \hbar\omega) = f(E_l) \xrightarrow{E_l \ll E_F} 1. \quad (14)$$

Rosei’s integrand $[1 - f(E)]$ is thus the $f(E_l) \rightarrow 1$ limit of ours; the bracket “gives the probability that a state with energy E in the upper band is empty,” in his words following Eq. (7). Retaining $f(E - \hbar\omega)$ explicitly — as we do — is the more general form, and is mandatory once the bands are driven out of equilibrium (elevated electronic temperature, pulsed excitation, or emission), where the initial-state occupation is no longer unity. At equilibrium absorption the two coincide identically.

6 Integration limits, line shape, and the L partner

The remaining content of Eqs. (6)–(8) is the domain of E and the E -dependence of $k_{\parallel}$. Solving (1) together with $\Omega_{lu} = \hbar\omega$ for the in-plane momentum (eliminate $k_{\perp}^2$) gives the dimensionally clean

$$k_{\parallel}^2 = \frac{1}{\mathcal{D}} \left[A_u (\hbar\omega - \mathcal{E}_g^X) - \bar{A} (E - \hbar\omega_{X_6^-}) \right] \quad (6')$$

the corrected form of Rosei's Eq. (6) (see section 7). Equation (6') shows $D_{l \rightarrow u} \propto k_{\parallel}^{-1} \propto (E_{\max} - E)^{-1/2}$: an integrable inverse-square-root edge in E at the $k_{\parallel} = 0$ end. The positivity $k_{\parallel}^2 \geq 0$ fixes the upper limit at $k_{\parallel} = 0$,

$$E_{\max} = \hbar\omega_{X_6^-} + \frac{A_u}{A} (\hbar\omega - \mathcal{E}_g^X) = \hbar\omega_{X_6^-} + \frac{m_{l\perp}}{m_{l\perp} + m_{u\perp}} (\hbar\omega - \mathcal{E}_g^X), \quad (15)$$

the corrected form of Rosei's Eq. (8). Because the saddle CEDS is *open*, the companion condition $k_{\perp}^2 \geq 0$ does not bound E from below; following Rosei the lower limit is the thermal floor $E_{\min} = -20 k_B T$, below which $[1 - f(E, T)]$ kills the integrand. The wide, thermally limited lower range makes the X EDJDOS look steplike in E , and the absorption rises smoothly through $\mathcal{E}_g^X = 1.94$ eV with a near-exponential sub-gap tail — the defining X signature.

6.1 The L point: the same machinery, one sign flipped

At L the upper band is an ellipsoid ($+k_{\parallel}^2$ curvature), so $\bar{B}_L = B_u + B_l$ and the curvature determinant takes the *difference* form $\mathcal{D}_L = A_u B_l - A_l B_u$. Every step of sections 3 and 4 goes through verbatim with $\mathcal{D} \rightarrow \mathcal{D}_L$. Now *both* positivity conditions bound E , so the closed ellipsoid yields a finite window

$$E_{\min}^L = \hbar\omega_{L_6^-} + \frac{B_u}{\bar{B}_L} (\hbar\omega - \mathcal{E}_g^L) \leq E \leq E_{\max}^L = \hbar\omega_{L_6^-} + \frac{A_u}{\bar{A}_L} (\hbar\omega - \mathcal{E}_g^L), \quad (16)$$

of width $\propto (\hbar\omega - \mathcal{E}_g^L)$. With $[1 - f] \approx \text{const}$ across the narrow near-onset window, $\mathcal{I}_{l \rightarrow u}^L \propto \int_{E_{\min}^L}^{E_{\max}^L} (E_{\max}^L - E)^{-1/2} dE = 2\sqrt{E_{\max}^L - E_{\min}^L} \propto \sqrt{\hbar\omega - \mathcal{E}_g^L}$ — the textbook M_0 inverse-square-root edge $\varepsilon_2 \propto \sqrt{\hbar\omega - \mathcal{E}_g^L}$, a sharp onset at $\mathcal{E}_g^L = 2.45$ eV with no sub-gap tail.

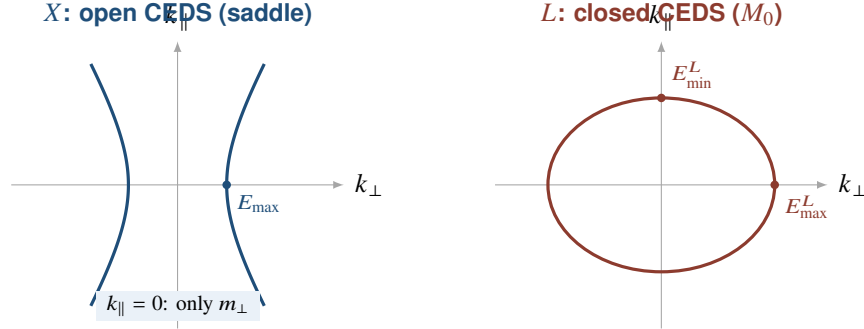


Figure 1: Constant-energy-difference surfaces $\Omega_{lu} = \hbar\omega$ in the $(k_{\perp}, k_{\parallel})$ plane. The upper integration limit $E_{\max}$ sits at $k_{\parallel} = 0$ on the $k_{\perp}$ axis, where the parallel curvatures B_u, B_l have dropped out of both constraints — hence $E_{\max}$ depends only on the *perpendicular* masses (section 7). At X the surface is an open hyperbola (sub-gap tail); at L it is a closed ellipse (sharp M_0 onset).

	X (saddle, M_1)	L (ellipsoid, M_0)
upper-band $k_{\parallel}^2$ sign	$-B_u$ (saddle)	$+B_u$ (ellipsoid)
CEDS topology	open hyperboloid	closed ellipsoid
curvature determinant	$\mathcal{D}_X = A_u B_l + A_l B_u$	$\mathcal{D}_L = A_u B_l - A_l B_u$
E range	$[-20k_B T, E_{\max}]$	$[E_{\min}^L, E_{\max}^L]$ finite
edge in E	steplike (box)	inverse- $\sqrt{}$
edge in $\hbar\omega$	smooth, sub-gap tail	sharp $\sqrt{\hbar\omega - \mathcal{E}_g^L}$
onset	1.94 eV	2.45 eV

7 Two printed equations of Rosei, examined

Asserting that a seminal paper contains a slip demands proof, not assertion. We examine Eqs. (6) and (8) as printed and show, on three independent grounds, what the corrected forms must be. We are careful to separate a *shorthand* (constants reshuffled, harmless) from a *transcription error* (wrong masses, consequential).

7.1 Eq. (6): a dimensional shorthand, correct masses

As printed, Rosei's Eq. (6) reads

$$k_{\parallel} = \left(\hbar\omega - \hbar\omega_{X_7^+} + \frac{\hbar^2}{2m_{l\perp}} (\hbar\omega_{X_6^-} - E) - \frac{\hbar^2}{2m_{u\perp}} E \right)^{1/2}. \quad (6, \text{ as printed})$$

Read literally in SI, the bracket adds an energy $(\hbar\omega - \hbar\omega_{X_7^+})$ to terms of the form $\frac{\hbar^2}{2m} \times \text{energy}$, which carry dimensions $(\text{J m}^2) \times \text{J} = \text{J}^2 \text{m}^2$; the sum cannot equal $k_{\parallel}^2 \sim \text{m}^{-2}$. For the bracket to have dimension m^{-2} the mass coefficients must read $2m/\hbar^2$, not $\hbar^2/2m$, and an overall $\mathcal{D}^{-1}$ must multiply the bracket. The dimensionally consistent statement is exactly (6'). *Crucially, the masses in Eq. (6) are the perpendicular ones, $m_{l\perp}$ and $m_{u\perp}$ — which is correct, since by (6') $k_{\parallel}^2$ depends only on A_u, A_l . We therefore read Eq. (6) as Rosei's own shorthand, in which the $2m/\hbar^2$ constants and the determinant are folded away into $\mathcal{F}_{l \rightarrow u}$ and the overall scale (consistent with the momenta being quoted "in units of $\pi/4a$ " in his Fig. 3). The spectral content — $k_{\parallel}$ linear-in- E , producing the $(E_{\max} - E)^{-1/2}$ edge — is unchanged.*

7.2 Eq. (8): parallel masses where perpendicular masses are required

As printed, Rosei's upper limit is his Eq. (8),

$$E_{\max} = \hbar\omega_{X_6^-} + (\hbar\omega_{X_7^+} + \hbar\omega_{X_6^-} - \hbar\omega) \frac{m_{u\parallel}}{m_{l\parallel} - m_{u\parallel}} = \hbar\omega_{X_6^-} - (\hbar\omega - \mathcal{E}_g^X) \frac{m_{u\parallel}}{m_{l\parallel} - m_{u\parallel}}, \quad (8, \text{ as printed})$$

which carries the *parallel* masses $m_{u\parallel}, m_{l\parallel}$. This cannot be correct, for three independent reasons.

(i) **Internal inconsistency with Eq. (6).** The upper limit is, by definition, the $k_{\parallel} = 0$ root of the very relation in Eq. (6). But Eq. (6) contains *only* the perpendicular masses $m_{l\perp}, m_{u\perp}$. A root of an equation in $\{m_{l\perp}, m_{u\perp}\}$ cannot introduce $\{m_{u\parallel}, m_{l\parallel}\}$. Equations (6) and (8) of the same paper are thus mutually inconsistent as printed; one of them has the wrong subscripts, and by (ii) it is (8).

(ii) **Geometric necessity.** Set $k_{\parallel} = 0$ directly in the two defining surfaces (1) and $\Omega_{lu} = \hbar\omega$:

$$E_{\max} = \hbar\omega_{X_6^-} + A_u k_{\perp}^2, \quad \hbar\omega = \mathcal{E}_g^X + \bar{A} k_{\perp}^2 \Rightarrow k_{\perp}^2 = \frac{\hbar\omega - \mathcal{E}_g^X}{\bar{A}}.$$

Neither equation contains B_u or B_l once $k_{\parallel} = 0$: the parallel curvatures have literally dropped out. Therefore (cf. fig. 1)

$$E_{\max} = \hbar\omega_{X_6^-} + \frac{A_u}{\bar{A}} (\hbar\omega - \mathcal{E}_g^X) = \hbar\omega_{X_6^-} + \frac{m_{l\perp}}{m_{l\perp} + m_{u\perp}} (\hbar\omega - \mathcal{E}_g^X),$$

a function of the *perpendicular* masses alone. The appearance of $m_{u\parallel}, m_{l\parallel}$ in Eq. (8) is structurally impossible for a $k_{\parallel} = 0$ edge.

(iii) **The printed coefficient is out of physical range.** The exact slope is the manifest fraction

$$\frac{A_u}{\bar{A}} = \frac{A_u}{A_u + A_l} \in (0, 1) \quad (\text{since } 0 < A_u < \bar{A}),$$

numerically $\frac{m_{l\perp}}{m_{l\perp} + m_{u\perp}} = \frac{0.19}{0.19 + 0.31} = 0.38$ with Rosei's X masses. The printed coefficient instead evaluates to

$$\frac{m_{u\parallel}}{m_{l\parallel} - m_{u\parallel}} = \frac{0.40}{0.15 - 0.40} = -1.6,$$

negative and of magnitude exceeding unity: it arises from the near-cancellation $m_{l\parallel} - m_{u\parallel}$ in the denominator and cannot represent a positive edge fraction in $(0, 1)$. A slope of -1.6 would place $E_{\max}$ above the photon energy and on the wrong side of $\hbar\omega_{X_6^-}$, contradicting fig. 1.

Corrected Eq. (8)

$$E_{\max} = \hbar\omega_{X_6^-} + \frac{A_u}{A} (\hbar\omega - \mathcal{E}_g^X) = \hbar\omega_{X_6^-} + \frac{m_{l\perp}}{m_{l\perp} + m_{u\perp}} (\hbar\omega - \mathcal{E}_g^X), \quad \frac{m_{l\perp}}{m_{l\perp} + m_{u\perp}} = 0.38.$$

The $\perp \leftrightarrow \parallel$ swap is the same class of transcription slip as the $\frac{\hbar^2}{2m} \leftrightarrow \frac{2m}{\hbar^2}$ shorthand in Eq. (6); here, however, it changes the masses that enter and so is consequential. It should be verified against the original print of Ref. 7; the OCR source used here reproduces the parallel subscripts, and the correction is forced regardless of which masses the print shows.

8 Assembly and conclusion

Collecting Steps 0–5, the two routes terminate in the same line integral, and the assembled dielectric function is Rosei's Eq. (9),

$$\varepsilon_2(\hbar\omega, T) = \frac{8\pi^2 e^2 \hbar^4}{3m^2 (\hbar\omega)^2} \left[|P_X|^2 N_X \mathcal{I}_{l \rightarrow u}^X(\hbar\omega, T) + |P_L|^2 N_L \mathcal{I}_{l \rightarrow u}^L(\hbar\omega, T) \right] + \varepsilon_2^{\text{Drude}}, \quad (17)$$

with $N_X = 6$, $N_L = 8$, the single fitted ratio $|P_X/P_L|^2 = 0.370$, and the constant $16\pi^3 \mathcal{F}_{l \rightarrow u} / \hbar^2$ of (11) absorbed into the strengths S_X, S_L of Eq. (10). To summarize the reconciliation:

- **Functional form:** identical. Both integrands are $\propto k_{\parallel}^{-1}(E, \omega)$ with the same $E_{\max}$ (15) and the same $k_{\parallel}^2(E)$ (6'); all spectral shape lives there.
- **Prefactor:** differs by the E, ω -independent constant $16\pi^3 \mathcal{F}_{l \rightarrow u} / \hbar^2$, whose dimension $2m/\hbar^2$ (12) is exactly what renders Rosei's $D_{l \rightarrow u}$ dimensionally complete. Absorbed into the overall scale.
- **Occupation:** Rosei's $[1 - f(E)]$ is our $f(E - \hbar\omega)[1 - f(E)]$ in the $f(E_l) \rightarrow 1$ (filled- d) limit; our explicit factor is the non-equilibrium-ready generalization needed for emission and pulsed excitation.
- **Printed equations:** Eq. (6) is a dimensional shorthand with correct (perpendicular) masses; Eq. (8) requires the perpendicular-mass correction (15), established on internal, geometric, and numerical grounds.

There was never a substantive disagreement — only Rosei's habit of folding the $2m/\hbar^2$ constants into $\mathcal{F}_{l \rightarrow u}$ and his normalization, together with the filled-valence-band approximation. The derivation given here reproduces Guerrisi–Rosei–Winsemius faithfully and exactly, while exposing the explicit, dimensionally complete, and non-equilibrium-ready form required for the emission and pulsed extensions of the model.

Source notes: Equivalence of My Absorption Integral and Rosei's.md and Interband Absorption at X and L (Rosei's Notation).md

*Primary reference: M. Guerrisi, R. Rosei & P. Winsemius, Phys. Rev. B **12**, 557 (1975) [Ref. 7]. Effective masses and gaps: Christensen & Seraphin, Phys. Rev. B **4**, 3321 (1971).*